

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277996

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

NISHIMOTO; Takuya

DISPLAY DEVICE

Abstract

According to an aspect, a display device includes: a front surface panel; a plurality of display units disposed at different positions on a back of the front surface panel; and a main frame supporting side parts of a first display unit and a second display unit adjacent to each other among the plurality of display units, and fixed to a back surface of the front surface panel.

Inventors: NISHIMOTO; Takuya (Tokyo, JP)

Applicant: Japan Display Inc. (Tokyo, JP)

Family ID: 80077557

Appl. No.: 19/210782

Filed: May 16, 2025

Foreign Application Priority Data

JP	2020-141638	Aug. 25, 2020
----	-------------	---------------

Related U.S. Application Data

parent US continuation 18753322 20240625 parent-grant-document US 12326623 child US 19210782

parent US continuation 18489626 20231018 parent-grant-document US 12339533 child US 18753322

parent US continuation 17992455 20221122 parent-grant-document US 11867992 child US 18489626

parent US continuation 17404112 20210817 parent-grant-document US 11543691 child US 17992455

Publication Classification

Int. Cl.: G02F1/1333 (20060101)

U.S. Cl.:

CPC G02F1/13332 (20210101); G02F1/133317 (20210101); G02F1/13336 (20130101); G02F1/133391 (20210101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 18/753,322, filed on Jun. 25, 2024, which is a continuation of U.S. application Ser. No. 18/489,626, filed on Oct. 18, 2023, which is a continuation of U.S. application Ser. No. 17/992,455, filed on Nov. 22, 2022, now U.S. Pat. No. 11,867,992, issued on Jan. 9, 2024, which is a continuation of U.S. application Ser. No. 17/404,112, filed on Aug. 17, 2021, now U.S. Pat. No. 11,543,691 issued on Jan. 3, 2023, which application claims the benefit of priority from Japanese Patent Application No. 2020-141638 filed on Aug. 25, 2020, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] What is disclosed herein relates to a display device.

2. Description of the Related Art

[0003] There is a transmissive display device including a backlight disposed on the back of a front surface panel and configured to display an image in a display region of the front surface panel by using light from the backlight (refer to Japanese Patent Application Laid-open Publication No. 2014-160218 (JP-A-2014-160218)). In JP-A-2014-160218, the front surface panel has the one display region and is supported by a frame.

[0004] A display device including a plurality of display regions on a front surface panel supported by a frame has been desired.

[0005] For the foregoing reasons, there is a need for a display device including a plurality of display regions on a front surface panel supported by a frame.

SUMMARY

[0006] According to an aspect, a display device includes: a front surface panel; a plurality of display units disposed at different positions on a back of the front surface panel; and a main frame supporting side parts of a first display unit and a second display unit adjacent to each other among the plurality of display units, and fixed to a back surface of the front surface panel.

[0007] According to an aspect, a display device includes: a front surface panel; a plurality of display units disposed on a back of the front surface panel and disposed at different positions; a first frame fixed to the back surface of the front surface panel and supporting a side part of a first display unit among the plurality of display units; and a second frame fixed to the back surface of the front surface panel, supporting a side part of a second display unit adjacent to the first display unit, and provided separately from the first frame.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a front view of a display device according to a first embodiment;

[0009] FIG. 2 is a cross-sectional view taken along line II-II in FIG. 1;

[0010] FIG. 3 is an enlarged cross-sectional view of an end part on an X1 side in FIG. 2;

[0011] FIG. 4 is an enlarged cross-sectional view of an end part on an X2 side in FIG. 2;
[0012] FIG. 5 is an enlarged cross-sectional view of a central part in an X direction in FIG. 2;
[0013] FIG. 6 is an oblique view of the cross-sectional view in FIG. 5;
[0014] FIG. 7 is a cross-sectional view of a display device according to a second embodiment and is a diagram corresponding to FIG. 5 according to the first embodiment;
[0015] FIG. 8 is an oblique view of the cross-sectional view in FIG. 7;
[0016] FIG. 9 is a cross-sectional view of a display device according to a third embodiment and is a diagram corresponding to FIG. 5 according to the first embodiment; and
[0017] FIG. 10 is an oblique view of the cross-sectional view in FIG. 9.

DETAILED DESCRIPTION

[0018] Aspects (embodiments) of the present disclosure will be described below in detail with reference to the accompanying drawings. Contents described below in the embodiments do not limit the present disclosure. Components described below include those that could be easily conceived by the skilled person in the art and those identical in effect. Components described below may be combined as appropriate.

[0019] The disclosure is merely exemplary, and any modification that could be easily conceived by the skilled person in the art as appropriate without departing from the gist of the disclosure is contained in the scope of the present disclosure. For clearer description, the drawings are schematically illustrated for the width, thickness, shape, and the like of each component as compared to an actual aspect in some cases, but the drawings are merely exemplary and do not limit interpretation of the present disclosure. In the present specification and drawings, any element same as that already described with reference to an already described drawing is denoted by the same reference sign, and detailed description thereof is omitted as appropriate in some cases.

First Embodiment

[0020] FIG. 1 is a front view of a display device according to a first embodiment. FIG. 2 is a cross-sectional view taken along line II-II in FIG. 1.

[0021] As illustrated in FIG. 1, a display device 1 according to the first embodiment is, for example, an in-car display device such as a car navigation system and has a horizontally long rectangular shape in which two light transmitting parts 21 and 22 are arranged in a right-left direction (X direction). Specifically, a cover glass (front surface panel) 2 is provided on a front surface of the display device 1. The light transmitting part 21 having a rectangular shape is provided on an X1 side of the cover glass 2, the light transmitting part 22 having a rectangular shape is provided on an X2 side thereof, and a light blocking part 23 is provided around the light transmitting parts 21 and 22. The light transmitting parts 21 and 22 are display regions of the display device 1, and the light blocking part 23 is a frame region of the display device 1. In the present disclosure, the number of light transmitting parts (display regions) is two or more but not limited to two and may be three or more.

[0022] As illustrated in FIG. 2, the display device 1 includes the cover glass (front surface panel) 2, a first display unit (display unit) 3, a second display unit (display unit) 4, and a main frame 5. The display units (the first display unit 3 and the second display unit 4) are disposed on the back (Z2 side) of the cover glass 2 at different positions with an interval therebetween in the X direction. In the present embodiment, the first display unit 3 and the second display unit 4 are provided. The first display unit 3 includes a liquid crystal display panel (LCD panel; also referred to as liquid crystal panel) 31 as an exemplary display panel, a backlight 32, a housing case 33, and a first coupling member 34 (refer to FIG. 3). The second display unit 4 includes a LCD panel 41 as an exemplary display panel, a backlight 42, a housing case 43, and a second coupling member 44 (refer to FIG. 4).

[0023] As illustrated in FIG. 2, the cover glass (front surface panel) 2 is disposed closest to a Z1 side in the display device 1. The cover glass 2 is a transparent glass member covering and

protecting display surfaces of the LCD panels **31** and **41** disposed on a back surface **2a** side (Z2 side). As described above with reference to FIG. 1, the cover glass **2** is provided with the light blocking part **23**. The light blocking part **23** is a region blocking light from the LCD panels **31** and **41**. The light blocking part **23** is formed by printing a light blocking member onto the cover glass **2** through performing screen-printing or the like. The light blocking member may be provided on any of a front surface **2b** and the back surface **2a** of the cover glass **2**.

[0024] The first display unit **3** will be described below. In the LCD panel (display panel) **31** illustrated in FIG. 2, a liquid crystal layer is disposed between two transparent substrates (refer to FIG. 6). The LCD panel **31** of the present embodiment is, for example, a liquid crystal display of a fringe field switching (FFS) type. A drive electrode (common electrode) and pixel electrodes formed for respective pixels in a matrix having a row-column configuration are stacked on one of the transparent substrates. At least one of the two transparent substrates is provided with a color filter including filters of R (red), G (green), B (blue), and the like arranged for the respective pixels. The LCD panel **31** has an opening formed at the pixel electrodes or the drive electrode (common electrode), and electric field (fringe electric field) leaking from the opening drives liquid crystals. The LCD panel **31** displays an image by switching between transmission and blocking of light at each pixel based on an image signal. The LCD panel **31** is not limited to the FFS type but may be a liquid crystal display of an in-plane switching (IPS) type, a twisted nematic (TN) type, an optically compensated bend (OCB; also referred to as optically compensated birefringence) type, or an electrically controlled birefringence (ECB) type.

[0025] The backlight **32** is disposed on the back surface side (Z2 side) of the LCD panel **31**. In other words, the backlight **32** is disposed opposite to the LCD panel **31** in a Z direction. The backlight **32** includes a light source (not illustrated) configured to output light and irradiates the LCD panel **31** with the light.

[0026] The housing case **33** is a box-shaped case made of resin or metal. The housing case **33** houses the backlight **32** and blocks light irradiated from the backlight **32** to prevent external leakage. The housing case **33** is preferably made of metal to obtain a heat releasing characteristic and the like. The first coupling member **34** will be described later in detail.

[0027] The main frame **5** includes a plate part **51** and leg parts **52** and **53** and is made of, for example, resin. The plate part **51** and the leg parts **52** and **53** are integrated. The plate part **51** extends along the back surface **2a** of the cover glass **2**. The plate part **51** is fixed to the back surface **2a** of the cover glass **2** with a double-sided adhesive tape **100**. The leg part **52** extends annularly along the side of the backlight **32**. The leg part **52** supports a side part of the first display unit **3**. The leg part **53** supports a side part of the second display unit **4** to be described later.

[0028] The second display unit **4** will be described below. In the LCD panel (display panel) **41** illustrated in FIG. 2, a liquid crystal layer is disposed between two transparent substrates (refer to FIG. 6). The structure and material of the LCD panel **41** of the present embodiment are the same as those of the LCD panel **31**.

[0029] The backlight **42** is disposed on the back surface side (Z2 side) of the LCD panel **41**. In other words, the backlight **42** is disposed opposite to the LCD panel **41** in the Z direction. The backlight **42** includes a light source (not illustrated) configured to output light and irradiates the LCD panel **41** with the light. The light source may be a light emitting diode (LED) or a fluorescence lamp.

[0030] The housing case **43** is a box-shaped case made of resin or metal. The housing case **43** houses the backlight **42** and blocks light irradiated from the backlight **42** to prevent external leakage. The housing case **43** is preferably made of metal to obtain a heat releasing characteristic and the like. The second coupling member **44** will be described later in detail.

[0031] The configuration of the display device **1** will be further specifically described with reference to FIGS. 3 to 6. FIG. 3 is an enlarged cross-sectional view of an end part on the X1 side in FIG. 2. FIG. 4 is an enlarged cross-sectional view of an end part on the X2 side in FIG. 2. FIG. 5

is an enlarged cross-sectional view of a central part in the X direction in FIG. 2. FIG. 6 is an oblique view of the cross-sectional view in FIG. 5.

[0032] As illustrated in FIG. 6, the backlight 32 included in the first display unit 3 includes optical sheets 321 and 322, a light guiding plate 323, a reflection sheet 324, and a heat releasing plate 325.

[0033] As illustrated in FIG. 6, the optical sheet 322 is stacked on the back (Z2 side) of the optical sheet 321, and the light guiding plate 323 is stacked on the back (Z2 side) of the optical sheet 322. The reflection sheet 324 is stacked on the back (Z2 side) of the light guiding plate 323, and the heat releasing plate 325 is stacked on the back (Z2 side) of the reflection sheet 324. The optical sheet 322 is, for example, a prism sheet and increases the directionality of light radiated from the front surface of the light guiding plate 323. The optical sheets 321 and 322 are a plurality of optical sheets included in the backlight and may be, for example, diffusive sheets, or polarization sheets. The optical sheets 321 and 322 may each have a single-layer structure or a multilayered structure. The reflection sheet 324 moderately diffuses light having the increased directionality. The reflection sheet 324 reflects light leaking from the light guiding plate 323, and thus the reflected light is incident onto the light guiding plate 323. The reflection sheet 324 is made of, for example, foamed polyethylene terephthalate (PET). The reflection sheet 324 may be, for example, a silver evaporation coating film, a multilayered reflection film, or a white PET. The reflection sheet 324 and the optical sheets 321 and 322 are provided as necessary and may be omitted. The heat releasing plate 325 may be made of, for example, aluminum or an aluminum alloy to obtain an increased heat releasing characteristic. The heat releasing plate 325 is provided as necessary and may be omitted. The heat releasing plate 325 may be integrally formed with the housing case 33. The LCD panel 31 is fixed to the back surface 2a of the cover glass 2 with a bonding layer 102.

[0034] As illustrated in FIGS. 3 and 5, the housing case 33 includes a bottom plate part 331 and a side plate part 332. The bottom plate part 331 is a flat plate member in the present embodiment. The bottom plate part 331 supports a back surface 32a of the backlight 32. The side plate part 332 extends from edges 333 and 334 of the bottom plate part 331 toward the front side (Z1 side). The edge 333 is positioned on the X2 side of the edge 334.

[0035] As illustrated in FIGS. 3 and 5, the first coupling member 34 includes a side plate part 341 and a front plate part (fifth plate part) 342. The side plate part 341 extends in the Z direction. The side plate part 341 is adjacent to and outside the side plate part 332 of the housing case 33 in the X direction. In other words, the side plate part 341 is formed in an annular shape surrounding the side plate part 332 on its inner periphery side. The front plate part 342 extends along the cover glass 2 to a position overlapping the backlight 32 in the Z direction. The front plate part 342 is in contact with a front surface 32b of the backlight 32. A gap (air layer) may be provided between the front plate part 342 and the front surface 32b of the backlight 32.

[0036] Thus, the backlight 32 is sandwiched between the front plate part 342 and the bottom plate part 331 in the Z direction. A protrusion (not illustrated) is provided at the side plate part 341 of the first coupling member 34 illustrated in FIG. 3, and a through-hole (not illustrated) is provided at the leg part 52 of the main frame 5. The protrusion of the side plate part 341 fits into the through-hole of the leg part 52, and thus the leg part 52 of the main frame 5 supports the side part of the first display unit 3. In addition, another through-hole is provided at the side plate part 341 of the first coupling member 34 illustrated in FIG. 3, and another protrusion (not illustrated) is provided at the side plate part 332 of the housing case 33. The protrusion of the side plate part 332 fits into the through-hole of the side plate part 341, and thus the first coupling member 34 supports a side part of the housing case 33. In this manner, the main frame 5, the first coupling member 34, and the housing case 33 are integrally fixed to each other.

[0037] As illustrated in FIG. 5, the main frame 5 has a cross-sectional shape of coupled two “T” figures at the central part in the X direction in FIG. 2. The leg parts 52 and 53 of the main frame 5 are adjacent to each other in the X direction. The leg parts 52 and 53 extend from the back surface of the plate part 51 toward the Z2 side. The leg part 52 is positioned on the X1 side of the leg part

53. An end part **51a** of the plate part **51** on the **X1** side is disposed to overlap the front plate part **342** of the first coupling member **34** when viewed in the **Z** direction. An elastic buffer material **101** is disposed between the end part **51a** and the front surface **32b** of the backlight **32**. In other words, the buffer material **101** is provided on the front side (**Z1** side) of the front plate part **342**, and the LCD panel **31** and the end part **51a** of the plate part **51** are provided on the front side (**Z1** side) of the buffer material **101**.

[0038] The second display unit **4** will be described below. As illustrated in FIG. **6**, the backlight **42** included in the second display unit **4** includes optical sheets **421** and **422**, a light guiding plate **423**, a reflection sheet **424**, and a heat releasing plate **425**.

[0039] As illustrated in FIG. **6**, the optical sheet **422** is stacked on the back (the **Z2** side) of the optical sheet **421**, and the light guiding plate **423** is stacked on the back (the **Z2** side) of the optical sheet **422**. The reflection sheet **424** is stacked on the back (the **Z2** side) of the light guiding plate **423**, and the heat releasing plate **425** is stacked on the back (the **Z2** side) of the reflection sheet **424**. The optical sheet **422** is, for example, a prism sheet and increases the directionality of light radiated from the front surface of the light guiding plate **423**. The optical sheets **421** and **422** are a plurality of optical sheets included in the backlight and may be, for example, diffusive sheets, or polarization sheets. The optical sheets **421** and **422** may each have a single-layer structure or may have a multilayered structure together.

[0040] The reflection sheet **424** moderately diffuses light having the increased directionality. The reflection sheet **424** reflects light leaking from the light guiding plate **423**, and thus the reflected light is incident onto the light guiding plate **423**. The reflection sheet **424** is made of, for example, foamed polyethylene terephthalate (PET). The reflection sheet **424** may be, for example, a silver evaporation coating film, a multilayered reflection film, or a white PET. The reflection sheet **424** and the optical sheets **421** and **422** are provided as necessary and may be omitted. The heat releasing plate **425** may be made of, for example, aluminum or an aluminum alloy to obtain an increased heat releasing characteristic. The heat releasing plate **425** is provided as necessary and may be omitted. The heat releasing plate **425** may be integrally formed with the housing case **43**. The LCD panel **41** is fixed to the back surface **2a** of the cover glass **2** with the bonding layer **102**.

[0041] As illustrated in FIGS. **4** and **5**, the housing case **43** includes a bottom plate part **431** and a side plate part **432**. The bottom plate part **431** is a flat plate member in the present embodiment. The bottom plate part **431** supports a back surface **42a** of the backlight **42**. The side plate part **432** extends from edges **433** and **434** of the bottom plate part **431** toward the front side (**Z1** side). The edge **433** is positioned on the **X1** side of the edge **434**.

[0042] As illustrated in FIGS. **4** and **5**, the second coupling member **44** includes a side plate part **441** and a front plate part (sixth plate part) **442**. The side plate part **441** extends in the **Z** direction. The side plate part **441** is adjacent to and outside the side plate part **432** of the housing case **43** in the **X** direction. In other words, the side plate part **441** is formed in an annular shape surrounding the side plate part **432** on its inner periphery side. The front plate part **442** extends along the cover glass **2** to a position overlapping the backlight **42** in the **Z** direction. The front plate part **442** is in contact with a front surface **42b** of the backlight **42**. A gap (air layer) may be provided between the front plate part **442** and the front surface **42b** of the backlight **42**.

[0043] Thus, the backlight **42** is sandwiched between the front plate part **442** and the bottom plate part **431** in the **Z** direction. A protrusion (not illustrated) is provided at the side plate part **441** of the second coupling member **44** illustrated in FIG. **4**, and a through-hole (not illustrated) is provided at the leg part **53** of the main frame **5**. The protrusion of the side plate part **441** fits into the through-hole of the leg part **53**, and thus the leg part **53** of the main frame **5** supports the side part of the second display unit **4**. In addition, another through-hole is provided at the side plate part **441** of the second coupling member **44** illustrated in FIG. **4**, and another protrusion (not illustrated) is provided at the side plate part **432** of the housing case **43**. The protrusion of the side plate part **432** fits into the through-hole of the side plate part **441**, and thus the second coupling member **44**

supports a side part of the housing case **43**. In this manner, the main frame **5**, the second coupling member **44**, and the housing case **43** are integrally fixed to each other.

[0044] As illustrated in FIG. 5, an end part **51b** of the plate part **51** on the X2 side is disposed to overlap the front plate part **442** of the second coupling member **44** when viewed in the Z direction. The elastic buffer material **101** is disposed between the end part **51b** and the front surface **42b** of the backlight **42**. In other words, the buffer material **101** is provided on the front side (Z1 side) of the front plate part **442**, and the LCD panel **31** and the end part **51b** of the plate part **51** are provided on the front side (Z1 side) of the buffer material **101**.

[0045] The display device **1** according to the first embodiment described above includes the cover glass (front surface panel) **2**, the plurality of display units disposed at different positions on the back of the cover glass **2**, and the main frame **5** supporting the side part of the first display unit **3** and the side part of the second display unit **4** and fixed to the back surface **2a** of the cover glass **2**. Thus, the display device **1** can be provided with a plurality (in the present embodiment, two) of display regions of the cover glass (front surface panel) **2** supported by the main frame **5**. In other words, in a display device according to JP-A-2014-160218 described above, a cover glass is supported by a frame but includes only one display region. However, in the present embodiment, the display device **1** including a plurality of display regions is provided.

[0046] The first display unit **3** includes the LCD panel (display panel) **31** fixed to the back surface **2a** of the cover glass **2**, the backlight **32** disposed on the back of the LCD panel **31** and configured to illuminate the LCD panel **31**, the housing case **33** housing therein the backlight **32**, and the first coupling member **34** coupling the housing case **33** and the main frame **5** to each other. In this manner, the housing case **33** housing therein the backlight **32** is coupled to the main frame **5** by the first coupling member **34**. The main frame **5** is fixed to the cover glass **2**. Therefore, the backlight **32**, the housing case **33**, the first coupling member **34**, and the cover glass **2** are united by the main frame **5**.

[0047] The second display unit **4** includes the LCD panel (display panel) **41** fixed to the back surface **2a** of the cover glass **2**, the backlight **42** disposed on the back of the LCD panel **41** and configured to illuminate the LCD panel **41**, the housing case **43** housing therein the backlight **42**, and the second coupling member **44** coupling the housing case **43** and the main frame **5** to each other. In this manner, the housing case **43** housing therein the backlight **42** is coupled to the main frame **5** by the second coupling member **44**. The main frame **5** is fixed to the cover glass **2**. Therefore, the backlight **42**, the housing case **43**, the second coupling member **44**, and the cover glass **2** are united by the main frame **5**.

[0048] In addition to the backlight **32**, the housing case **33**, the first coupling member **34**, and the cover glass **2** described above, the backlight **42**, the housing case **43**, the second coupling member **44**, and the cover glass **2** are united by the main frame **5**. Since the components of the entire display device **1** are united by the main frame **5** in this manner, the display device **1** can be compactly assembled with a simple configuration.

[0049] The first coupling member **34** includes the front plate part (fifth plate part) **342** extending along the cover glass **2** to a position overlapping the backlight **32**, and the backlight **32** is sandwiched between the front plate part **342** and the housing case **33**. Thus, the backlight **32** can be housed in the housing case **33** with a simple configuration.

[0050] The second coupling member **44** includes the front plate part (sixth plate part) **442** extending along the cover glass **2** to a position overlapping the backlight **42**, and the backlight **42** is sandwiched between the front plate part **442** and the housing case **43**. Thus, the backlight **42** can be housed in the housing case **43** with a simple configuration.

Second Embodiment

[0051] A display device **1A** according to a second embodiment will be described below. Any portion having the same configuration as that of the display device **1** according to the first embodiment described above is denoted by the same reference sign and description thereof is

omitted. Specifically, a frame of the display device **1A** of the second embodiment is different from that of the display device **1** of the first embodiment and thus will be described below in detail.

[0052] FIG. **7** is a cross-sectional view of the display device according to the second embodiment and is a diagram corresponding to FIG. **5** according to the first embodiment. FIG. **8** is an oblique view of the cross-sectional view in FIG. **7**.

[0053] As illustrated in FIGS. **7** and **8**, the display device **1A** according to the second embodiment includes a first frame **105** and a second frame **110**.

[0054] The first frame **105** has a T-shaped cross section. Specifically, the first frame **105** includes a plate part **106** and a leg part **107**. The plate part **106** and the leg part **107** are integrated and made of, for example, resin. The plate part **106** extends along the back surface **2a** of the cover glass **2**. The plate part **106** is fixed to the back surface **2a** of the cover glass **2** by the double-sided adhesive tape **100**.

[0055] An end part **106a** of the plate part **106** on the X1 side is disposed to overlap the front plate part **342** of the first coupling member **34** when viewed in the Z direction. The elastic buffer material **101** is disposed between the end part **106a** and the front surface **32b** of the backlight **32**. In other words, the buffer material **101** is provided on the front side (Z1 side) of the front plate part **342**, and the LCD panel **31** and the end part **106a** of the plate part **106** are provided on the front side (Z1 side) of the buffer material **101**.

[0056] The leg part **107** extends annularly along the side of the backlight **32**. The leg part **107** extends from the back surface of the plate part **106** to the Z2 side and supports the side part of the first display unit **3**. Specifically, a protrusion (not illustrated) is provided at the side plate part **341** of the first coupling member **34** illustrated in FIGS. **7** and **8**, and a through-hole (not illustrated) is provided at the leg part **107** of the first frame **105**. The protrusion of the side plate part **341** fits into the through-hole of the leg part **107**, and thus the leg part **107** of the first frame **105** supports the side part of the first display unit **3**. In addition, another through-hole is provided at the side plate part **341** of the first coupling member **34** illustrated in FIGS. **7** and **8**, and another protrusion (not illustrated) is provided at the side plate part **332** of the housing case **33**. The protrusion of the side plate part **332** fits into the through-hole of the side plate part **341**, and thus the first coupling member **34** supports the side part of the housing case **33**. In this manner, the first frame **105**, the first coupling member **34**, and the housing case **33** are integrally fixed to each other.

[0057] The second frame **110** has a T-shaped cross section. Specifically, the second frame **110** includes a plate part **111** and a leg part **112**. The plate part **111** and the leg part **112** are integrated and made of, for example, resin. The plate part **111** extends along the back surface **2a** of the cover glass **2**. The plate part **111** is fixed to the back surface **2a** of the cover glass **2** by the double-sided adhesive tape **100**.

[0058] An end part **111a** of the plate part **111** on the X2 side is disposed to overlap the front plate part **442** of the second coupling member **44**. The elastic buffer material **101** is disposed between the end part **111a** and the front surface **32b** of the backlight **42**. In other words, the buffer material **101** is provided on the front side (Z1 side) of the front plate part **442**, and the LCD panel **41** and the end part **111a** of the plate part **111** are provided on the front side (Z1 side) of the buffer material **101**.

[0059] The leg part **112** extends annularly along the side of the backlight **32**. The leg part **112** extends from the back surface of the plate part **111** to the Z2 side and supports the side part of the second display unit **4**. Specifically, a protrusion (not illustrated) is provided at the side plate part **441** of the second coupling member **44** illustrated in FIGS. **7** and **8**, and a through-hole (not illustrated) is provided at the leg part **112** of the second frame **110**. The protrusion of the side plate part **441** fits into the through-hole of the leg part **112**, and thus the leg part **112** of the second frame **110** supports the side part of the second display unit **4**. In addition, another through-hole is provided at the side plate part **441** of the second coupling member **44** illustrated in FIGS. **7** and **8**, and another protrusion (not illustrated) is provided at the side plate part **432** of the housing case **43**. The protrusion of the side plate part **432** fits into the through-hole of the side plate part **441**, and

thus the second coupling member **44** supports the side part of the housing case **43**. In this manner, the second frame **110**, the second coupling member **44**, and the housing case **43** are integrally fixed to each other. An end edge **106b** of the plate part **106** of the first frame **105** and an end edge **111b** of the plate part **111** of the second frame **110** are separated from each other in the X direction. In other words, the end edge **106b** and the end edge **111b** are opposite to each other in the X direction with a gap therebetween.

[0060] According to the second embodiment described above, advantages described below are obtained. The display device **1A** includes the cover glass (front surface panel) **2**, a plurality of display units disposed on the back of the cover glass **2** and disposed at different positions, the first frame **105** fixed to the back surface **2a** of the cover glass **2** and supporting the side part of the first display unit **3** among the plurality of display units, and the second frame **110** fixed to the back surface **2a** of the cover glass **2**, supporting the side part of the second display unit **4** adjacent to the first display unit **3**, and provided separately from the first frame **105**.

[0061] Thus, such an advantage is obtained that the display device **1A** can be more easily assembled than in the first embodiment while the relative position of the first display unit **3** with respect to the cover glass **2** and the relative position of the second display unit **4** with respect to the cover glass **2** are appropriately held. This advantage will be briefly described below.

[0062] As a premise, both the relative position of the first display unit **3** with respect to the cover glass **2** and the relative position of the second display unit **4** with respect to the cover glass **2** need to be set to appropriate positions.

[0063] First, in a case of the first embodiment, the main frame **5** is attached to the first display unit **3** and the cover glass **2** after the relative position of first display unit **3** with respect to the cover glass **2** is attached appropriately.

[0064] Subsequently, the second display unit **4** is assembled to the main frame **5**. When the variation amount of shaping dimensions of the main frame **5** is larger than a predetermined amount, the error of the relative position of the second display unit **4** with respect to the cover glass **2** exceeds an allowable amount in some cases. In such a case, work is required to remove the cover glass **2** from the first display unit **3**, correct the position of the first display unit **3**, and then attach the first display unit **3** to the cover glass **2**.

[0065] On the other hand, in a case of the second embodiment, the relative position of the first display unit **3** with respect to the cover glass **2** is corrected first, and then the first frame **105** is attached to the first display unit **3**.

[0066] Subsequently, the relative position of the second display unit **4** with respect to the cover glass **2** is corrected, and then the second frame **110** is attached to the second display unit **4**.

[0067] In this manner, in the second embodiment, the first frame **105** and the second frame **110** are individually assembled to the first display unit **3** and the second display unit **4**, respectively. Thus, the relative position of the first display unit **3** with respect to the cover glass **2** and the relative position of the second display unit **4** with respect to the cover glass **2** can be easily set to appropriate positions.

[0068] The first display unit **3** includes the LCD panel (display panel) **31** fixed to the back surface **2a** of the cover glass **2**, the backlight **32** disposed on the back of the LCD panel **31** and configured to illuminate the LCD panel **31**, the housing case **33** housing therein the backlight **32**, and the first coupling member **34** coupling the housing case **33** and the first frame **105** to each other. In this manner, the housing case **33** housing therein the backlight **32** is coupled to the main frame **5** by the first coupling member **34**. The first frame **105** is fixed to the cover glass **2**. Therefore, the backlight **32**, the housing case **33**, the first coupling member **34**, and the cover glass **2** are united by the first frame **105**, and thus the display device **1A** can be compactly assembled with a simple configuration.

[0069] The second display unit **4** includes the LCD panel (display panel) **41** fixed to the back surface **2a** of the cover glass **2**, the backlight **42** disposed on the back of the LCD panel **41** and

configured to illuminate the LCD panel **41**, the housing case **43** housing therein the backlight **42**, and the second coupling member **44** coupling the housing case **43** and the second frame **110** to each other. The second frame **110** is fixed to the cover glass **2**. Therefore, the backlight **42**, the housing case **43**, the second coupling member **44**, and the cover glass **2** are united by the second frame **110**, and thus the display device **1A** can be compactly assembled with a simple configuration.

Third Embodiment

[0070] A display device **1B** according to a third embodiment will be described below. Any portion having the same configuration as that of the display device **1** according to the first embodiment described above is denoted by the same reference sign and description thereof is omitted.

Specifically, a frame of the display device **1B** of the third embodiment is different from that of the display device **1** of the first embodiment and thus will be described below in detail.

[0071] FIG. **9** is a cross-sectional view of the display device according to the third embodiment and is a diagram corresponding to FIG. **5** according to the first embodiment. FIG. **10** is an oblique view of the cross-sectional view in FIG. **9**.

[0072] As illustrated in FIGS. **9** and **10**, a main frame **205** according to the third embodiment includes a first subframe **210** and a second subframe **220**. The first subframe **210** and the second subframe **220** are integrally fixed to each other by the double-sided adhesive tape **100**.

[0073] The first subframe **210** has a T-shaped cross section. Specifically, the first subframe **210** includes a first plate part **230** and a first leg part **240**. The first plate part **230** and the first leg part **240** are integrated and made of, for example, resin. The first plate part **230** extends along the back surface **2a** of the cover glass **2**. The first plate part **230** is fixed to the back surface **2a** of the cover glass **2** by the double-sided adhesive tape **100**.

[0074] An end part **230a** of the first plate part **230** on the **X1** side is disposed to overlap the front plate part **342** of the first coupling member **34** when viewed in the **Z** direction. The elastic buffer material **101** is disposed between the end part **230a** and the front surface **32b** of the backlight **32**. In other words, the buffer material **101** is provided on the front side (**Z1** side) of the front plate part **342**, and the LCD panel **31** and the end part **230a** of the first plate part **230** are provided on the front side (**Z1** side) of the buffer material **101**.

[0075] The first leg part **240** extends annularly along the side of the backlight **32**. The first leg part **240** extends from the back surface of the first plate part **230** to the **Z2** side and supports the side part of the first display unit **3**. Specifically, a protrusion (not illustrated) is provided at the side plate part **341** of the first coupling member **34** illustrated in FIGS. **9** and **10**, and a through-hole (not illustrated) is provided at the first leg part **240** of the first subframe **210**. The protrusion of the side plate part **341** fits into the through-hole of the first leg part **240**, and thus the first leg part **240** of the first subframe **210** supports the side part of the first display unit **3**. In addition, another through-hole is provided at the side plate part **341** of the first coupling member **34** illustrated in FIGS. **9** and **10**, and a protrusion (not illustrated) is provided at the side plate part **332** of the housing case **33**. The protrusion of the side plate part **332** fits into the through-hole of the side plate part **341**, and thus the first coupling member **34** supports the side part of the housing case **33**. In this manner, the first subframe **210**, the first coupling member **34**, and the housing case **33** are integrally fixed to each other.

[0076] The second subframe **220** has a cross-sectional shape of combined two “L” figures.

Specifically, the second subframe **220** includes a second plate part **250**, a second leg part **260**, a third plate part **270**, a third leg part **280**, and a fourth plate part **290**. The second plate part **250** and the second leg part **260** form an L-shaped cross section, and the third plate part **270** and the third leg part **280** form an L-shaped cross section. The second plate part **250**, the second leg part **260**, the third plate part **270**, the third leg part **280**, and the fourth plate part **290** are integrated.

[0077] The second plate part **250** and the second leg part **260** have an L-shaped cross section in combination. The second plate part **250** extends along the back surface **2a** of the cover glass **2**. The second plate part **250** is fixed to the back surface **2a** of the cover glass **2** by the double-sided

adhesive tape **100**. An end part **250b** of the second plate part **250** on the X2 side is disposed to overlap the front plate part **442** of the second coupling member **44**. The elastic buffer material **101** is disposed between the end part **250b** and the front surface **42b** of the backlight **42**. In other words, the buffer material **101** is provided on the front side (Z1 side) of the front plate part **442**, and the LCD panel **41** and the end part **250b** of the second plate part **250** are provided on the front side (Z1 side) of the buffer material **101**.

[0078] The second leg part **260** extends annularly along the side of the backlight **42**. The second leg part **260** extends from the back surface of the second plate part **250** to the Z2 side and supports the side part of the second display unit **4**. Specifically, a protrusion (not illustrated) is provided at the side plate part **441** of the second coupling member **44** illustrated in FIGS. **9** and **10**, and a through-hole (not illustrated) is provided at the second leg part **260** of the second subframe **220**. The protrusion of the side plate part **441** fits into the through-hole of the second leg part **260**, and thus the second leg part **260** of the second subframe **220** supports the side part of the second display unit **4**. In addition, another through-hole is provided at the side plate part **441** of the second coupling member **44** illustrated in FIGS. **9** and **10**, and another protrusion (not illustrated) is provided at the side plate part **432** of the housing case **43**. The protrusion of the side plate part **432** fits into the through-hole of the side plate part **441**, and thus the second coupling member **44** supports the side part of the housing case **43**. In this manner, the second subframe **220**, the second coupling member **44**, and the housing case **43** are integrally fixed to each other.

[0079] The third plate part **270** extends from a central part of the second leg part **260** in the Z direction to the X1 side. In other words, the third plate part **270** extends from the second leg part **260** along the back surface of the first plate part **230**. The third plate part **270** is fixed to the back surface of an end part **230b** of the first plate part **230** by the double-sided adhesive tape **100**. Thus, the end part **230b** of the first plate part **230** on the X2 side is opposite to the third plate part **270** in the Z direction, and the end part **230b** and the third plate part **270** are bonded to each other by the double-sided adhesive tape **100**.

[0080] The third leg part **280** extends from an end part **270a** of the third plate part **270** on the X1 side to the Z2 side. In other words, the third leg part **280** extends from the third plate part **270** along the first leg part **240** of the first subframe **210**. The third plate part **270** and the third leg part **280** have an L-shaped cross section in combination.

[0081] The fourth plate part **290** has a rectangular shape and extends so as to be orthogonal to (intersect) the third plate part **270**. The fourth plate part **290** couples the second leg part **260**, the third plate part **270**, and the third leg part **280**. The fourth plate part **290** also extends so as to be orthogonal to (intersect) the second leg part **260** and the third leg part **280**. A plurality of the fourth plate parts **290** are provided at intervals in a Y direction as illustrated in FIG. **10**.

[0082] According to the third embodiment described above, advantages described below are obtained. The main frame **205** includes the first subframe **210** and the second subframe **220** fixed to the first subframe **210**, the first subframe **210** supports the side part of the first display unit **3**, and the second subframe **220** supports the side part of the second display unit **4**. With this configuration, the second subframe **220** can be fixed to the first subframe **210** in a state where the position of the second subframe **220** with respect to the first subframe **210** is changed as appropriate. Thus, the relative position of the first display unit **3** with respect to the cover glass **2** and the relative position of the second display unit **4** with respect to the cover glass **2** can be easily set to appropriate positions.

[0083] The first subframe **210** includes the first plate part **230** fixed to the back surface of the cover glass **2**, and the first leg part **240** extending from the first plate part **230** along the side part of the first display unit **3** and supporting the side part. The second subframe **220** includes the second plate part **250** fixed to the back surface **2a** of the cover glass **2**, the second leg part **260** extending from the second plate part **250** along the side part of the second display unit **4** and supporting the side part, and the third plate part **270** extending from the second leg part **260** along the back surface of

the first plate part **230** and fixed to the back surface.

[0084] Thus, the cover glass **2** can be supported from the back by the first leg part **240** and the second leg part **260**. Moreover, in order to fix the third plate part **270** to the back surface of the first plate part **230**, the first subframe **210** and the second subframe **220** can be fixed in a state where the relative positions of the first subframe **210** and the second subframe **220** are appropriately adjusted.

[0085] The second subframe **220** includes the third leg part **280** extending from the third plate part **270** along the first leg part **240**. Thus, the first plate part **230** of the first subframe **210** can be supported by the third leg part **280**.

[0086] The second subframe **220** includes the fourth plate part **290** extending so as to be orthogonal to (intersect) the third plate part **270** and coupling the second leg part **260**, the third plate part **270**, and the third leg part **280** to one another, whereby stiffness of the second leg part **260**, the third plate part **270**, and the third leg part **280** improves.

[0087] Preferable embodiments of the present disclosure are described above, but the present disclosure is not limited to such embodiments. Contents disclosed in the embodiments are merely exemplary, and various kinds of modifications are possible without departing from the scope of the present disclosure. Any modification performed as appropriate without departing from the scope of the present disclosure belongs to the technical scope of the present disclosure. At least one of various kinds of omission, replacement, and change of components can be performed without departing from the scope of the embodiments and modification described above. For example, the number of display regions is set to be two in the embodiments, but may be three or more. A bonding agent may be used instead of the double-sided adhesive tape **100** used for bonding of the cover glass **2** and the main frame **5**, bonding of the cover glass **2** and each of the first frame **105** and the second frame **110**, bonding of the cover glass **2** and each of the first subframe **210** and the second subframe **220**.

Claims

1. A display device comprising: a cover glass; a plurality of display panels including a first display panel fixed to a back surface of the cover glass; a first subframe fixed to the back surface of the cover glass; and a second subframe fixed to the back surface of the cover glass, wherein the first subframe includes: a first plate part fixed to the back surface of the cover glass; and a first leg part extending from the first plate part along a side part of the first display panel, the second subframe includes: a second plate part fixed to the back surface of the cover glass and separated from the first plate part; a second leg part extending from the second plate part; and a third plate part extending from the second leg part along a back surface of the first plate part.
2. The display device according to claim 1, further comprising a second display panel disposed with a gap from the first display panel in a first direction, to the back surface of the cover glass, wherein the second leg part extends from the second plate part along a side part of the second display panel.
3. The display device according to claim 2, wherein a length of the first plate part is longer than a length of the second plate part in the first direction.
4. The display device according to claim 1, further comprising: a first backlight disposed on a back of the first display panel; a first housing case housing therein the first backlight; and a first coupling member coupling the first housing case and the first subframe to each other, wherein the first coupling member includes a fifth plate part extending along the cover glass to a position overlapping the first backlight, and the first backlight is sandwiched between the fifth plate part and the first housing case.
5. The display device according to claim 1, further comprising a plurality of double-sided adhesive tapes including a first tape, a second tape, and a third tape, wherein the first plate part of the first subframe is fixed to the cover glass with the first tape, the second plate part of the second subframe

is fixed to the cover glass with the second tape, and the third plate part of the second subframe is fixed to the first plate part of the first subframe with the third tape.
